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Solar System Missions: Psyche

Journey to a Metal World

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Introduction

Psyche is a mission to the strangest place in the asteroid belt: a metal world. Its target, the asteroid 16 Psyche, is the largest of the rare metallic asteroids, a body some two hundred kilometers across that may be the exposed core of a small planet that never finished forming, a planetesimal stripped of its rocky shell in the solar system's violent first hundred million years. If the hypothesis holds, the asteroid is the only way any scientist will ever see a planetary core close up, since the cores of Earth, Venus, Mars, and Mercury lie sealed forever beneath thousands of kilometers of rock, and the mission NASA launched toward it in October 2023 is a visit to the inside of a world.

The mission is a Discovery-class project, the trade's mid-sized class, led by a university for the first time in the program's history, built on a commercial satellite bus, and propelled across the solar system by the gentle thrust of electric engines that breathe sunlight for fuel. Its instruments are few and sharp: cameras to map the metal world, spectrometers to taste its chemistry, a magnetometer to test the memory of an ancient dynamo, and the spacecraft's own radio to weigh the body as it orbits. Along for the ride flies a technology demonstration that may matter as much as the science, a laser communications system testing whether light can carry data across interplanetary distances.

This book tells the mission whole: the discovery of the asteroid in 1852 and the naming that gave it the Greek word for soul, the long argument about what the metal world is made of, the mission's difficult birth and its delayed launch, the electric cruise and the Martian flyby, the instruments and their questions, and the meaning of a journey to the one place in the solar system that resembles the hidden heart of every rocky planet, the whole campaign, told in the order it happened, the machine, the metal world, and the core of the argument about both.

A word on dates before the story starts: this account follows the project's public calendar exactly, the world found in 1852, the program's choice landing in 2017, the liftoff of October 2023, the swing past Mars in 2026, and the rendezvous planned for 2029, every date taken from the mission's own literature. Where the calendar is still blank, at the mapping orbits and the final verdict, the account admits it plainly, and the reader who wants the raw material can find it in the project's repositories, signposted through the later chapters, the record kept, audited, and retold here, gently, once.

The mission's identity should be fixed early, since the name carries a double: Psyche the spacecraft is not the mythological romance the contaminated predecessor of this volume carried, a tale the rebuild replaced entirely, but the asteroid mission, named for its target, the sixteenth asteroid, whose own name came from the Greek word for soul, the name's lineage, stated once here, and the mission, understood by its destination, the metal world, and the heart it may be, the name, earned, by the target, and honored, by the flight.

The book then walks the story in its natural order: the early chapters tell how the world was found and what it is; the middle chapters carry the physics, the instruments read as the craft's five senses, the measuring campaign already underway, and the prospectors' question that follows metal wherever it is named; the later chapters gather what the mission may find, what could still go wrong, what comes after arrival, and the trivia the journey has already collected; and the reader who wants the whole flight in a single sentence can take this one: a spacecraft launched in 2023 toward the one object that may be a planet's bare heart, riding whispering electric engines, carrying a handful of instruments and a laser telephone, and due to settle a century-old argument in 2029, at the length the metal world has waited for.

Chapter 1: The Discovery of Psyche

The discovery story of Psyche begins in Naples, on March 17, 1852, when Annibale de Gasparis, the director of the observatory there and one of the nineteenth century's most prolific asteroid hunters, found a moving point in the constellation's charts, the sixteenth asteroid, a body the numbering called 16 Psyche, the discovery story's first fact, the metal world found by a man who spent his career finding worlds, and the reader who studies the asteroid's early records finds the era's modest means: a refractor, a chart, and patience, the nineteenth century's whole toolkit, and the metal world, found with it, the first entry in the campaign.

The naming followed the era's convention: the mythological pattern the catalog had settled into, the sixteenth asteroid named for the figure the Greek world made the soul's personification, the name given in the era's ordinary way, and the asteroid carried a word for soul into the catalog, the naming, inherited, from the myth's vocabulary, and the mission that came a hundred and seventy years later inherited the name's second life, the discovery story's linguistic middle, the name, borrowed, and the meaning, kept.

What the mission truly discovered, and what this chapter recounts, is the strangeness of the target itself: the photometry and spectroscopy of the twentieth century sorted Psyche into the metallic class, the M-type, the reflectance odd and the radar signature bright, the body suspected of being metal rather than rock, the discovery that made the asteroid interesting to the modern trade, the realization that the sixteenth asteroid might be a fragment of a planet's heart, and the closing insight of the chapter: the discovery story of Psyche is the story of a name and a substance converging, a soul, found, in the metal, and a mission, proposed eventually, to meet it.

The discovery chapter's spare note is the asteroid's early size in the era's arithmetic: the body estimated at some two hundred kilometers across, the largest of its class, the size that made it findable in 1852 and interesting in every century after, the note the reader should carry as the metal world's scale, a mountain range's worth of metal, possibly, and the trade's only such candidate, the size, measured roughly then, and refined since, the discovery story's physical anchor, and the mission's, eventual target, dated from the Naples charts.

The chapter also marks the hunt's context: de Gasparis was among the century's great asteroid hunters, the man who found the belt's early members one by one, the sixth, the seventh... his catalog's long list, and Psyche, his later finds, the reader who studies the era's astronomy sees the asteroid belt being discovered, body by body, in the decades between the eight and the hundred, and this metal world, among the earliest hundred, the era's patient survey, kept, and the metal world, found by it, the discovery story's institutional frame, and the observatory's, quiet monument.

The discovery chapter's last credit is the era's catalog itself: the numbering that made the asteroid the sixteenth, the symbol the century drew for it, a butterfly and a star, the notation the trade retired and the collectors remember, and the reader who consults the asteroid's early literature finds the body's whole biography in the nineteenth century's records, the name, given, and the symbol, drawn, and the metal, unrecognized, for another hundred years, the credit this chapter closes on, the catalog, begun, and the soul, entered, into the ledger of worlds, which is every asteroid's true first chapter, and this one's, especially, given the name's later fulfillment, in the mission's destination, and the trade's, long memory, kept, permanently.

The chapter's closing ledger is the metal world's own biography across two centuries of records: the nineteenth century's charts and the twentieth century's photometry, the body tracked through every opposition since 1852, the orbit refined until the occultations and the radar could take their readings, the register that every targeted asteroid earns, the biographical patience, kept, and the target's reliability, proven, by a hundred and seventy years of continuous observation, which is the trade's quietest prerequisite, and this mission's, inherited, from the Naples charts, and honored by the destination, for good and, effectively, forever.

The number deserves a line, because the asteroid's own name begins with one. The sixteen in its designation is the nineteenth century's ledger mark, the sixteenth minor planet to be given a number, the counting being the era's filing system: each new world entered the ledger in order of discovery, the number permanent, the name following, and the metal world has carried its sixteenth-place mark since before the American Civil War, the designation being older than most nations' flags; and the number's persistence is the astronomers' own honesty, a name being a courtesy and a number being a fact, the metal world being, in the catalogs, still the sixteenth, the ledger having been kept continuously for a hundred and seventy years, the chapter's opening paragraph being itself a ledger entry, the newest in a long bookkeeping that began with the first dot found wandering.